

Capstone Topic Approval Form

The purpose of this document is to help you clearly explain your capstone topic, project scope, and timeline. Identify each of the following areas so you will have a complete and realistic overview of your project. Your course instructor cannot approve your project topic without this information.

Student Name: Dustin Keate

Student ID: [REDACTED]

Capstone Project Name: Optimistic Analysis: Evaluating S&P 500 Analyst Accuracy

Project Topic: Analysis of stock analyst sentiment bias against target revision directional accuracy.

Research Question: Do financial analyst firms that exhibit a higher proportion of positive stock ratings (hereon "optimistic") have better accuracy and success over 90-day and 365-day horizons?

Hypothesis: Analyst firms classified as optimistic show a higher rate of directional accuracy and forward returns compared to the remaining analyst population.

Context: In the financial services sector, equity research analysts serve as vital information sources, providing ratings and price targets to guide participants in financial markets. However, there is a frequent adage among retail investment (and one that I follow as well), "time in the markets beats timing the markets." In a MSCI blog post, Wickermasinghe describes this advice and provides in depth analysis to support it (2026).

Understanding whether an analyst's optimistic outlook results in actual outperformance could be informative to a retail investor pursuing advice or a wealth management specialist looking to create recommendations for a client. If highly optimistic analysts consistently outperform the broader market, their targets may serve as reliable indicators for portfolio growth.

By analyzing analyst revisions across the S&P 500 index, this project will seek to provide investors with a data-driven context to evaluate analyst performance based on historical optimism.

Data: I will use a Kaggle dataset, "S&P 500 Analyst Rating and Price Target Accuracy" (Urad, 2026). Also, I will also look at market downturn period data in my exploratory analysis (Yardeni Research, 2024).

The dataset contains 164,231 individual analyst predictions spanning a period from December 2011 through June 2026.

The dataset was provided to Kaggle by user Urad. It was provided under the CC BY-NC-SA 4.0 license. This license allows use as long as I provide attribution, use it for a non-commercial purpose, and share my project after it is complete.
<https://creativecommons.org/licenses/by-nc-sa/4.0/>



Data Gathering: I will gather data from Kaggle and readily available, published market research. I may perform my own analysis to define and find downturns in the S&P 500 using an existing market API.

[CHANGE] Data Analytics Tools and Techniques: I will use VS Code as my development environment, GitHub as my version control and backup solution, and a local JupyterLab environment for analysis. In my analysis, I will use Python, including common data analysis packages such as pandas, numpy, scipy, sklearn, and matplotlib, among others. I will perform an exploratory data analysis and generate data and charts using standard data analysis techniques used in this course of study.

Since I am seeking to identify correlation between analyst optimism and prediction accuracy, I will use correlation testing in my analysis. I currently plan to use a rank correlation such as Spearman. To identify statistical significance, I will calculate p-values where appropriate.

I may elect to apply a model from sklearn to create an evidence-based prediction generator if I am able to identify a statistically significant correlation.

Justification of Tools/Techniques: VS Code is a common IDE and supports multiple languages and tools. GitHub is the industry standard in git version management. Python and its extensive standard libraries will be able to transform the dataset into a useful and human understandable format. Additionally, my application of appropriate testing and models will meet course requirements.

Application Type, if applicable (select one):

- ☐ mobile
- ☐ web
- ☒ stand-alone

Programming/Development Language(s), if applicable: Python.

Operating System(s)/Platform(s), if applicable: Windows 11 running WSL.

Database Management System, if applicable: N/A.

Project Outcomes: My outcomes and deliverables for this project include:

- A GitHub project showing project development over time, implementing industry best practices learned during this program.
- A Jupyter Notebook File showing a thoroughly documented and reproducible Python pipeline showing the full development cycle from CSV ingestion to visual product production.
- A final written report that explains methodology and analysis used to resolve my hypothesis.

Projected Project End Date: 8/15/2026

Sources:

Urad. (2026). S&P 500 Analyst Rating and Price Target Accuracy [Data set]. Kaggle.
<https://www.kaggle.com/datasets/uradkr/s-and-p-500-analyst-rating-and-price-target-accuracy>



Wickremasinghe, J. (2026 April 28). Time in the markets beats timing the markets. MSCI.
<https://www.msci.com/research-and-insights/blog-post/time-in-the-markets-beats-timing-the-markets>

Yardeni Research. (2024 January 21). Stock market historical tables: Bull & bear markets.
<https://old.yardeni.com/wp-content/uploads/BullBearTables.pdf>

Human Subjects or Proprietary Information

Does your project involve the potential use of human subjects? (Y/N): No.

Does your project involve the potential use of proprietary company information? (Y/N): No.

STUDENT SIGNATURE



By signing and submitting this form, you acknowledge that any cost associated with the development and execution of your data analytics solution will be your (the student) responsibility.

